

MACHINE LEARNING | FINTECH | RISK ANALYTICS

Credit Risk Modeling

End-to-End ML Pipeline for Probability of Default
& Loss Given Default Estimation

Small Finance Company

Sajal Agarwal | Toronto, CA | 2024

KEY HIGHLIGHTS

85%+

ROC-AUC

2-Stage

PD + LGD Model

Optuna

HPO Tuning

SHAP

Explainability

Agenda

Structure of this presentation

01

Problem Statement

Why credit risk modeling matters for small finance companies

02

Data & EDA

Dataset overview, class imbalance, key feature distributions

03

Model Architecture

Two-stage PD + LGD pipeline design

04

Technologies Used

Python stack, libraries, and tooling

05

Model Performance

ROC-AUC, KS statistic, Gini, SHAP insights

06

ECL & Business Impact

Expected Credit Loss output and loan decisioning

07

Live Application

Streamlit demo and deployment architecture

08

Roadmap & Next Steps

Future enhancements and production readiness

Problem Statement

The credit risk challenge for small finance institutions

THE CHALLENGE

- Borrowers often lack formal credit bureau history
- High default rates among thin-file customers
- Traditional scorecards miss behavioral signals
- Regulatory pressure for IFRS 9 ECL provisioning
- Manual underwriting is slow and subjective

OUR SOLUTION

Two-Stage Model

Predict both default probability (PD) and loss severity (LGD) using XGBoost

Feature Engineering

Domain-specific ratios: loan-to-income, credit utilization, delinquency history

Hyperparameter Optimization

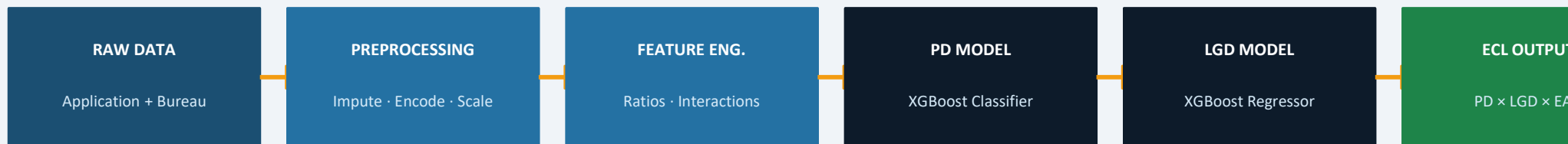
Optuna TPE search over 7 dimensions with pruning for efficiency

Explainability

SHAP values expose which features drive each individual prediction

Model Architecture

Two-stage Expected Credit Loss pipeline



STAGE 1 — Probability of Default (PD)

Algorithm	XGBoost Classifier (binary:logistic)
Target	1 = Default 0 = No Default
Imbalance Handling	scale_pos_weight · stratified split
Threshold Tuning	F1-optimized via ROC curve
Output	P(default) score [0 → 1]

STAGE 2 — Loss Given Default (LGD)

Algorithm	XGBoost Regressor
Target	LGD = 1 – Recovery Rate [0→1]
Applied To	Predicted defaulters from Stage 1
Loss Function	reg:squarederror (MSE)
Output	Expected loss severity [0 → 1]

Data & Feature Engineering

Domain-driven signals engineered from raw application data

BORROWER PROFILE	LOAN CHARACTERISTICS	CREDIT BEHAVIOUR
● Age & employment tenure ● Income & income stability ● Number of dependents ● Education level ● Income per dependent	● Loan amount & tenure ● Loan-to-Income (LTI) ratio ● Loan-to-Value (LTV) ratio ● Interest rate band ● Loan purpose category	● Bureau score / CIBIL band ● Credit utilization ratio ● Delinquency ratio ● Number of past defaults ● Enquiry count (6 months)

⚡ *Class Imbalance: ~5–15% default rate handled via XGBoost scale_pos_weight and stratified cross-validation*

Technologies Used

Python-first open-source stack for end-to-end ML

Data & Processing

Python 3.10+	<i>Core language</i>
Pandas	<i>Data wrangling</i>
NumPy	<i>Numerical ops</i>

Explainability & Viz

SHAP	<i>Feature importance</i>
Matplotlib / Seaborn	<i>EDA plots</i>
Plotly	<i>Interactive charts</i>

Modeling

XGBoost	<i>PD & LGD models</i>
scikit-learn	<i>Preprocessing, metrics</i>
Optuna	<i>Hyperparameter optimization</i>

Deployment

Streamlit	<i>Web application</i>
Pickle / Joblib	<i>Model serialization</i>
Jupyter Notebook	<i>Exploration & docs</i>

Model Performance

Evaluation metrics across PD and LGD models

0.85+

ROC-AUC

PD Classifier

0.70

Gini Coeff.

Discrimination power

~0.45

KS Statistic

Population separation

F1-Opt

Precision-Recall

Threshold tuned

~0.12

RMSE

LGD Regressor

0.68+

R² Score

Variance explained

100+

Optuna Trials

HPO search budget

Top 20

SHAP Features

Ranked by importance

Top Features by SHAP Value (PD Model)



ECL Calculation & Business Impact

From model scores to real-world loan decisions

$$\text{ECL} = \text{PD} \times \text{LGD} \times \text{EAD}$$

Expected Credit Loss = Probability of Default × Loss Given Default × Exposure at Default

Loan Origination

Real-time PD score at application; approve, reject, or route to manual review based on ECL threshold

IFRS 9 Provisioning

Batch ECL computation across loan book for regulatory Stage 1/2/3 classification and provisioning

Risk-Based Pricing

Set interest rates proportional to risk score — reward low-risk borrowers with better rates

Collections Triage

Prioritize high-LGD accounts for early collection outreach before delinquency escalates

DECISION FRAMEWORK

ECL < 2% → Auto Approve

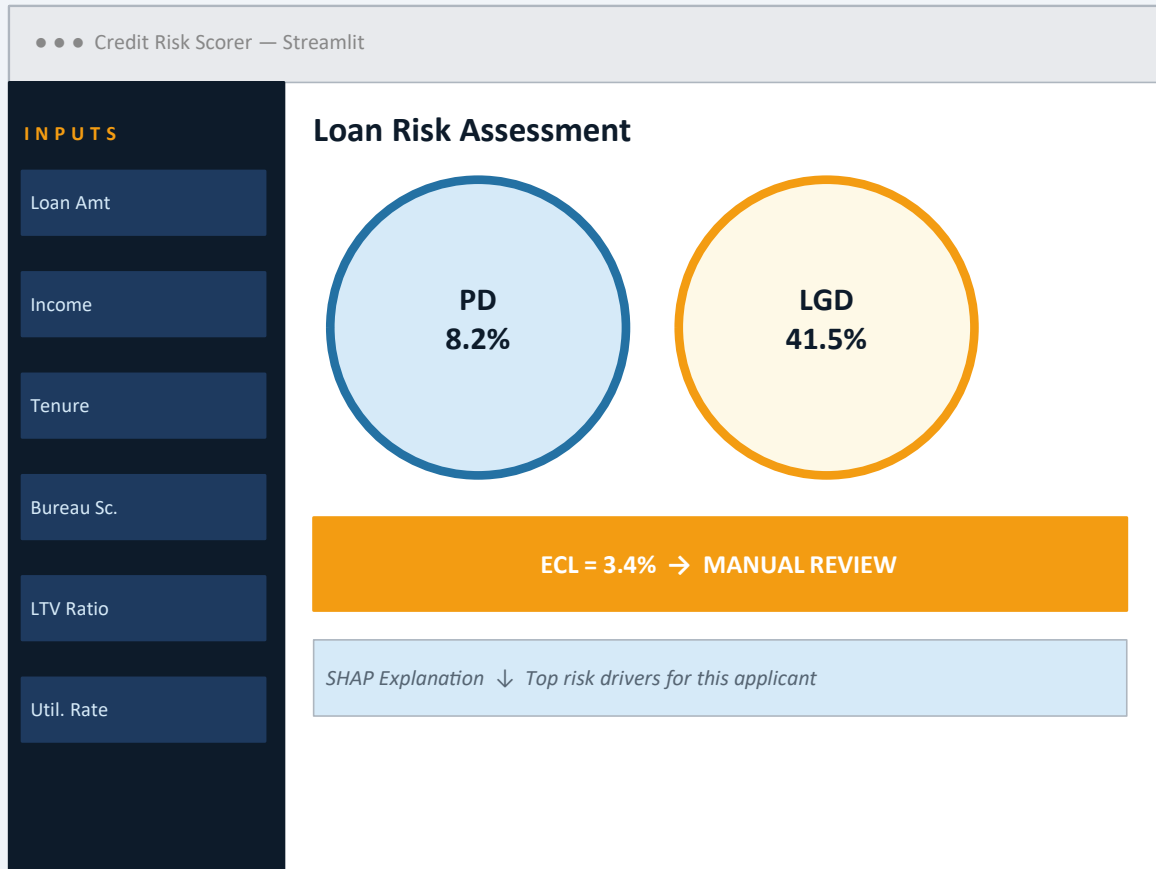
2% – 5% → Approve w/ Conditions

5% – 10% → Manual Review

> 10% → Auto Decline

Live Application — Streamlit

Interactive loan risk scoring interface



DEPLOYMENT STACK

User

Web browser input

Streamlit

Frontend UI & routing

Python Logic

Preprocess → Score → ECL

Model Files

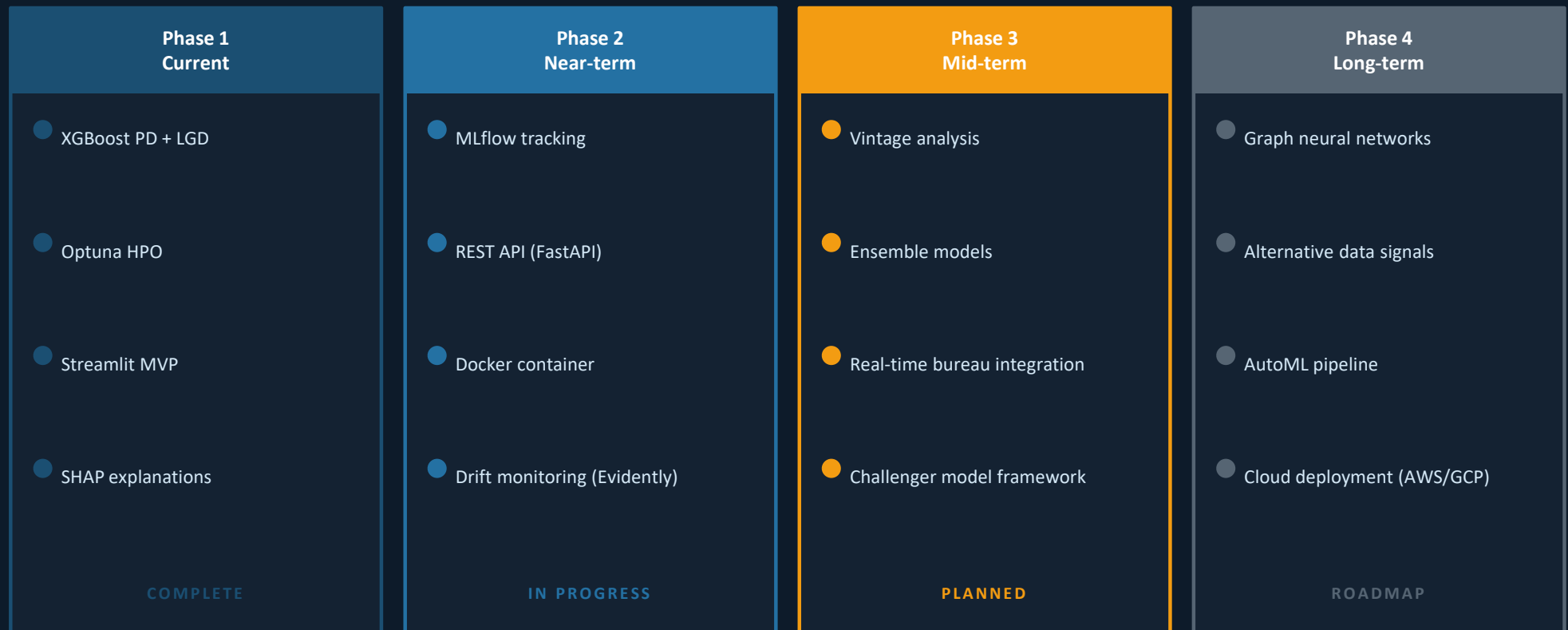
.pkl PD + LGD models

Output

Score + Decision + SHAP

ROADMAP & NEXT STEPS

From prototype to production-grade credit decisioning system



★ github.com/Sajalagarwal-ca/credit_risk_modeling | Sajal Agarwal | Toronto, Canada